1 · Symbols & cash-flow timing

S_0, S_t, S_T: underlying now, at t, at expiry; K: strike; T: years; r: continuous domestic risk-free rate. c, p: European call/put; C, P: American.

$x^+ = \max(x, 0)$. PV of A at T: Ae^{-rT}; future value: Ae^{rT}. D: PV of known dividends; q: continuous dividend yield; r_f: foreign rate.

Payoff = expiry cash flow. **Profit** = payoff minus cost (same time basis). Premium at 0: subtract premium for unfinanced convention, or premium e^{rT} at expiry if financed.

2 · Contracts & four positions

Forward: obligation; long $S_T - K$, short $K - S_T$. Futures: standardized, daily mark-to-market. Option: buyer's right, writer's obligation; premium upfront.

European: exercise at T only. American: may exercise earlier. Hedger reduces exposure; speculator takes risk; arbitrage buys cheap/sells expensive equivalent cash flows.

Long position	Payoff	BE*
Call	$(S_T - K)^+$	$K + c_0$
Put	$(K - S_T)^+$	$K - p_0$

Short = negative long payoff/profit. *BE ignores financing. Long option max loss = premium; call upside unbounded; put payoff at most K.

ITM call $S > K$; ITM put $S < K$; ATM $S = K$. ITM ≠ profitable. Intrinsic = exercise payoff; time value = price − intrinsic.

Example: $K = 50, c_0 = 4, S_T = 62$: payoff 12, profit 8. At 48: payoff 0, profit −4.

3 · Trading & value drivers

Multiply per-unit amounts by contract multiplier (lecture stock convention: 100). Class: call/put on underlying; series: same strike/expiry. Close long by selling same series; close short by buying it.

Bid 4/ask 4.50: spread .50; midpoint 4.25; immediate 100-share round trip $50 before fees.

Mark-to-market: observed quote; mark-to-model: model inputs. Margin is collateral. Initial deposit; maintenance threshold; margin call restores required balance. Futures/writers can face obligations; fully paid option buyer loses at most premium.

Input ↑	Call	Put
S / K	+ / −	− / +
σ	+	+
r / dividends	+ / −	− / +

Longer expiry: American value cannot fall; European not unconditional. $C \geq c$, $P \geq p$.

4 · Bounds: no dividends

Frictionless trade, shorting and borrowing/lending at r; same underlying/strike/expiry for comparisons.

$(S_0 - Ke^{-rT})^+ \leq c \leq S_0$
$(Ke^{-rT} - S_0)^+ \leq p \leq Ke^{-rT}$

American: $(S_0 - K)^+ \leq C \leq S_0$; $(K - S_0)^+ \leq P \leq K$ for $r \geq 0$.

Proof pattern: call + bond paying K dominates stock; put + stock dominates bond. Buy underpriced dominating portfolio, sell dominated one.

5 · Parity, dividends & exercise

$c + Ke^{-rT} = p + S_0$ (European, no dividends). Both sides pay $\max(S_T, K)$.

$c = p + S_0 - Ke^{-rT}$
$p = c + Ke^{-rT} - S_0$
$S_0 = c - p + Ke^{-rT}$.

Arbitrage: sell expensive side, buy cheap side; initial receipt, terminal cancellation. Negative bond term = borrow.

Known dividends: D = their PV.
$c + D + Ke^{-rT} = p + S_0$
$c \geq (S_0 - D - Ke^{-rT})^+$
$p \geq (D + Ke^{-rT} - S_0)^+$.

American bounds ($r \geq 0$):
$S_0 - K \leq C - P \leq S_0 - Ke^{-rT}$
with dividends, lower bound $S_0 - D - K$.

No-dividend call, $r \geq 0$: $C = c$; do not exercise early (lose optionality, pay strike sooner). Put may exercise deep ITM to receive K earlier. Dividends may favor call exercise just before ex-dividend. Always compare continuation.

At a cash dividend: $S_{after} = S_{before} - D_{cash}$. Cash amount ≠ PV ≠ continuous yield. A yield CRR tree is not an exact cash-dividend tree.

6 · Strategy quick reference

Same expiry unless noted. $K_1 < K_2 < K_3$; A = net debit. Profit below ignores financing. Add signed legs, split at strikes, subtract A.

Position	Payoff / use
Covered call $S - c(K)$	$\min(S_T, K)$ Income, capped upside
Protective put $S + p(K)$	$\max(S_T, K)$ Floor on stock value
Bull call $c(K_1) - c(K_2)$	$\min((S_T - K_1)^+, K_2 - K_1)$ Moderately bullish
Bear put $p(K_2) - p(K_1)$	$\min((K_2 - S_T)^+, K_2 - K_1)$ Moderately bearish

Spreads: max loss A, max gain $K_2 - K_1 - A$; bull BE $K_1 + A$, bear BE $K_2 - A$.

European box: bull call + bear put. Constant payoff $K_2 - K_1$, value $(K_2 - K_1)e^{-rT}$. American early exercise complicates replication.

Butterfly: $c(K_1) - 2c(K_2) + c(K_3)$, equal spacing. Peak payoff $K_2 - K_1$ at K_2; zero outside wings. Low-movement view.

Example: strikes 90, 100, 110, premiums 14.2, 8.1, 4.3: $A = 2.3$, max profit 7.7, BEs 92.3, 107.7.

Straddle: $c(K) + p(K)$, payoff $|S_T - K|$; BEs $K \pm A$.

Strangle: $p(K_1) + c(K_2)$; payoff $(K_1 - S_T)^+ + (S_T - K_2)^+$; BEs $K_1 - A, K_2 + A$. Lower cost, larger move needed.

Strip: call + 2 puts (down bias): $(S_T - K)^+ + 2(K - S_T)^+$. **Strap:** 2 calls + put (up bias): $2(S_T - K)^+ + (K - S_T)^+$.

Calendar: short near, long far, same strike. Far leg retains time value at near expiry. **Diagonal:** different strikes and expiries.

Principal protection: budget $A \to$ bond Ae^{-rT} + call budget $A(1 - e^{-rT})$, assuming nonnegative budget and risk-free bond.

7 · One-step replication · no yield

$S_u = S_0u$, $S_d = S_0d$; f_u, f_d option values; Δ shares per option unit.

$\Delta = (f_u - f_d)/(S_u - S_d)$
$H = \Delta S_u - f_u = \Delta S_d - f_d$
$f_0 = \Delta S_0 - He^{-r\Delta t}$.

Long Δ shares − one option is riskless. To hedge a short of M option units, hold $M\Delta$ shares (negative means short shares).

Example: $S_0 = 20$, up 22, down 18, $K = 21, r = .04, \Delta t = .25$. Payoffs 1, 0; $\Delta = .25$, $H = 4.5$; $f_0 = S - 4.5e^{-.01} = .5448$.

8 · Risk-neutral pricing

$p = (a - d)/(u - d)$; require $d < a < u$ so $0 < p < 1$.
$f = e^{-r\Delta t}[pf_u + (1 - p)f_d]$

Underlying	Growth a
No-dividend stock	$e^{r\Delta t}$
Yield stock / index	$e^{(r-q)\Delta t}$
FX (domestic/foreign)	$e^{(r-r_f)\Delta t}$
Futures	1

Discount at **domestic r in all cases**. FX foreign rate acts as yield. Non-income forward $F_0 = S_0e^{rT}$.

Under Q: $E^Q[S_{next} \mid S] = Sa$. Physical stock probability $p^* = (e^{\mu\Delta t} - d)/(u - d)$. p prices; p^* forecasts. Non-dividend: $dS/S = \mu dt + \sigma dW$ becomes $rdt + \sigma dW^Q$ under Q; same volatility.

9 · CRR and multi-step tree

N steps; i time index; j up moves.
$\Delta t = T/N$, $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$
$S_{i,j} = S_0u^jd^{i-j}$.

CRR matches local diffusion variance as step shrinks. Exact relative variance $p(1 - p)(u - d)^2 = a(u + d) - ud - a^2$. Convergence need not be monotonic.

Two-step European:
$f_0 = e^{-2r\Delta t}[p^2f_{uu}$
$+2p(1 - p)f_{ud} + (1 - p)^2f_{dd}]$.

Terminal values: calls $(S_{N,j} - K)^+$; puts $(K - S_{N,j})^+$.

European: discounted continuation.
American: $f_{i,j} = \max(V, \text{exercise})$ at every earlier node, including root. Recompute delta as stock/time changes.

10 · Backward induction pseudocode

```
set dt,u,d,a,p,disc=exp(-r*dt)
check d < a < u
for j=0..N:
  v[j]=payoff(S0*u^j*d^(N-j))
for i=N-1 down to 0:
  for j=0..i:  # ascending
    V=disc*((1-p)*v[j]+p*v[j+1])
    v[j]=V
    if American:
      v[j]=max(V,payoff(S0*u^j*d^(i-j)))
return v[0]
```

Time $O(N^2)$; rolling memory $O(N)$; storing whole tree $O(N^2)$.

Check: L6 two-step call root .9497 (L7 p.13's .9334 is inconsistent). L6 put: European 4.1923, American 5.0894; exercise at low node $S = 40$.

11 · Stochastic notation & rules

W_t (also z_t in slides): Wiener level; ΔW: increment; ε, U, V: standard normal shocks. $N(m, v)$ = mean m, **variance** v. Φ = standard normal CDF.

$a(X,t)$: drift; $b(X,t)$: diffusion; b^2 = variance rate. This a differs from binomial growth. F_t: information at t; E_t, Var_t: conditional moments.

Drift units: value/time; diffusion: value/$\sqrt{\text{time}}$; variance rate: value²/time. Use years consistently (month 1/12, quarter 1/4).

$E[\sum X_i] = \sum E[X_i]$ always.
$Var(\sum X_i) = \sum Var(X_i)$
$+2\sum_{i<j} Cov(X_i, X_j)$.
$Var(aX + bY) = a^2\,Var\,X$
$+b^2\,Var\,Y + 2ab\,Cov(X, Y)$.

Independent ⇒ uncorrelated (finite moments). Converse only with extra conditions, e.g. joint Gaussianity.

12 · Standard Wiener / Brownian

$W_0 = 0$; continuous paths; independent increments on disjoint intervals; $W_t - W_s \sim N(0, t-s)$.

$E[W_T] = 0$, $Var(W_T) = T$
$SD(W_T) = \sqrt{T}$
$\Delta W = \sqrt{\Delta t}\varepsilon$, fresh $\varepsilon \sim N(0,1)$ each step.

Variances add; SDs do not. Quarter time → quarter variance, half SD. Continuous path ≠ differentiable path.

Markov: conditional future given present does not need earlier history. Does not imply independent levels. Time/state may be discrete or continuous.

Example: $\Delta t = .01 \Rightarrow \Delta W = .1\varepsilon$. $W_5 - W_2 \sim N(0, 3)$. $P(W_1 > 1.5) = 1 - \Phi(1.5) = .0668$.

13 · Generalized Wiener: exact

Constant a, b: $dX = a\,dt + b\,dW$
$X_T = X_0 + aT + bW_T$
$X_T \sim N(X_0 + aT, b^2T)$
$\Delta X \sim N(a\Delta t, b^2\Delta t)$
$SD(X_T) = |b|\sqrt{T}$.

Exact finite-step update:
$X_{next} = X + a\Delta t + b\sqrt{\Delta t}\varepsilon$.

For $T > 0, b \neq 0$:
$P(X_T < L) = \Phi\left(\frac{L - X_0 - aT}{|b|\sqrt{T}}\right)$.

Q12: $X_0 = 30, a = 8, b = 12$. Drift 8, variance rate 144; $X_1 \sim N(38, 144)$. Half-year mean 34, two-year SD $12\sqrt{2}$. $P(X_1 < 25) = \Phi(-13/12) = .1393$.

14 · Itô process: local moments

$dX = a(X,t)dt + b(X,t)dW$
$E_t[\Delta X] \approx a(X_t, t)\Delta t$
$Var_t(\Delta X) \approx b(X_t, t)^2\Delta t$
$SD_t(\Delta X) \approx |b(X_t, t)|\sqrt{\Delta t}$.

Freeze coefficients at current state. Exact for constant coefficients; generally approximate otherwise.

Euler: $X_{next} = X + a(X,t)\Delta t$
$+b(X,t)\sqrt{\Delta t}\varepsilon$.

15 · Mean reversion

$dX = \kappa(\theta - X)dt + \eta dW$
$\kappa > 0$: speed; θ: target; η: diffusion. Drift positive below target, negative above. Variance rate η^2.

Q13: $\kappa = .6, \theta = 50, \eta = 8$. At $X = 40, 50, 65$: drift $6, 0, -9$. At $40, \Delta t = .1$: local mean change .6, variance 6.4.

16 · GBM: dollars vs percentages

$dS = \mu S dt + \sigma S dW$
$dS/S = \mu dt + \sigma dW$.

Local change	Mean	SD
Proportional	$\mu\Delta t$	$\sigma\sqrt{\Delta t}$
Dollars	$\mu S\Delta t$	$\sigma S\sqrt{\Delta t}$

Dollar variance $\sigma^2 S^2 \Delta t$. Drift coefficient μS; diffusion σS; variance rate $(\sigma S)^2$. μ is annual expected proportional drift, σ annualized volatility.

Local normal approximation ≠ exact long-horizon GBM return distribution. If $\sigma = 0$: exact deterministic $S_t = S_0 e^{\mu t}$.

Q15: $S = 120, \mu = .09, \sigma = .24, \Delta t = 1/12$. Proportional mean .0075, SD .069282; dollar mean .9, SD 8.3138.

17 · Euler simulation

$S_{i+1} = S_i(1 + \mu\Delta t + \sigma\sqrt{\Delta t}\varepsilon_i)$

```
dt=T/N; S=S0
for i=0..N-1:
  draw fresh epsilon ~ N(0,1)
  S=S*(1+mu*dt+sigma*sqrt(dt)*epsilon)
  store S if a path is needed
```

Time $O(N)$ per path; $O(1)$ rolling state, $O(N)$ stored path. M paths: $O(MN)$ time.

Use updated S_i, not always S_0. Independent shocks ≠ independent prices. Euler can give negative prices; smaller steps reduce discretization error, not randomness.

Q16: $S_0 = 100, \mu = .06, \sigma = .2$, monthly shocks $.5, -1.2, .8$: prices 103.3868, 96.7408, 101.6928 (verify without rounding between steps).

18 · Correlated increments

ρ = correlation, $-1 \le \rho \le 1$.
$Cov(\Delta W_1, \Delta W_2) = \rho\Delta t$
$Cov_t(\Delta X_1, \Delta X_2) \approx \rho b_1 b_2 \Delta t$
$Cov_t(\Delta S_1, \Delta S_2) \approx \rho\sigma_1\sigma_2 S_1 S_2 \Delta t$.

Constant coefficients: exact increment covariance. Local correlation = ρ for positive coefficients; exact for constant coefficients. Finite-horizon state-dependent changes need not have correlation ρ.

Q18: $a_1 = 3, b_1 = 5, a_2 = 2, b_2 = 7, \rho = .65, \Delta t = .25$: means .75, .5; variances 6.25, 12.25; covariance 5.6875; correlation .65.

19 · Generate correlated normals

Independent $U, V \sim N(0,1)$:
$\varepsilon_1 = U, \quad \varepsilon_2 = \rho U + \sqrt{1-\rho^2}V$.
$E[\varepsilon_i] = 0$; $Var(\varepsilon_2) = \rho^2 + (1-\rho^2) = 1$; $Cov(\varepsilon_1, \varepsilon_2) = \rho$.

```
each step: draw fresh independent U,V
e1=U; e2=rho*U+sqrt(1-rho^2)*V
update each stock with its own e
```

Same U within pair; fresh pair across time. $\rho = 0$: independent; $\rho = \pm 1$: perfect dependence.

20 · Brownian covariance identities

$Cov(W_s, W_t) = \min(s, t)$
$Corr(W_s, W_t) = \min(s,t)/\sqrt{st}$ for $s, t > 0$.

Forward intervals $a \le b, c \le d$:
$Cov(W_b - W_a, W_d - W_c)$
$= \max(0, \min(b,d) - \max(a,c))$
= overlap length.

General linear combinations:
$Cov(\sum_i a_i W_{t_i}, \sum_j b_j W_{s_j})$
$= \sum_{i,j} a_i b_j \min(t_i, s_j)$.

Linear Brownian combinations are jointly Gaussian: covariance 0 $\iff$ independence. For nonlinear functions, use actual independence properties; zero covariance alone is insufficient.

21 · All nine dependence cases

Pair	Cov / result
W_2, $W_4 - W_2$	0 · independent
W_2, W_4	2 · dependent
$W_3 - W_1$, $W_5 - W_3$	0 · independent
$W_3 - W_1$, $W_4 - W_2$	1 · dependent
$W_5 - W_2$, $W_2 - W_1$	0 · independent
$W_4 + W_2$, $W_4 - W_2$	2 · dependent
$W_3 - 2W_2 + W_1$, $W_3 - W_1$	0 · independent
W_2^2, $W_4 - W_2$	Independent*
$W_4 - 2W_2$, W_2	−2 · dependent

Case g: let $X = W_2 - W_1$, $Y = W_3 - W_2$, iid $N(0,1)$. Pair $Y - X, Y + X$ has covariance $1 - 1 = 0$, jointly normal.

*Case h: squaring a variable independent of the future increment preserves independence; the square is not Gaussian.

22 · Exam calculation patterns

Bounds: compute PV strike first. **Parity:** same K, T, check dividends and style. **Profit:** subtract premiums only after combining payoffs.

Tree: time units → u, d, a, p → check p → stock nodes → terminal payoff → backward discount → American exercise comparison → delta.

SDE: identify a, b → substitute current state → variance rate b^2 → multiply by Δt → SD square root. Mark exact/local.

Q14: $a = 4 + .2X, b = .15X$ at 80: $a = 20, b = 12$, monthly mean 1.6667, variance 12; Euler $80 + 20/12 + \sqrt{12}\varepsilon$.

Q17: $S = 90, \mu = .07, \sigma = .25$: quarterly mean 1.575, variance 126.5625; monthly .525, 42.1875.

Q20: $S_1 = 100, S_2 = 80$, $\mu_1 = .08, \mu_2 = .05$, $\sigma_1 = .2, \sigma_2 = .3$, $\rho = .5$, month, local/Euler moments: means .6667, .3333; SDs 5.7735, 6.9282; covariance 20.

23 · Common traps / scope

Variance is SD squared. Brownian level is not increment. ΔW scales with $\sqrt{\Delta t}$, drift with Δt. Uncorrelated need not mean independent. Risk-neutral probability is not physical probability. American value uses max, not just expectation.

Sources: supplied L1–9, Brownian supplement and Suggested Exercises 1. Full website: all 20 questions / 97 subparts + lecture exercises. This two-page sheet is a selected reference, not every worked solution.

L2's referenced margin spreadsheet was not supplied. L7 p.13 root value corrected against L6. No unprovided advanced pricing chapters included.